

Distributed Safety-Critical Optimal Flocking Control Algorithm with Feasibility Enhancement of High-Order Control Barrier Function

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Abstract—This paper proposes a distributed safety-critical optimal flocking control algorithm for the multi-agent system. In general, safety (i.e., collision avoidance) could be guaranteed by the use of a control barrier function (CBF) as constraints in the optimal control problem. However, the existence of conflicting safety constraints from multiple agents could lead to a feasibility problem (i.e., the absence of a feasible control command) in distributed optimal flocking control algorithms. To deal with this issue, we propose two effective feasibility enhancement approaches. First, we develop a high-order CBF associated with the collision risk to alleviate the burden of distributed safety constraints on agents with a high risk of collision. Second, to enhance feasibility, a relaxation method is used to enlarge the allowable control space of agents. In particular, to decrease the number of optimization variables in the relaxation method, a selective strategy is adopted. This involves applying the relaxation method only to specific CBF based on the decision variable related to the overlap of admissible control spaces among agents. Numerical simulations demonstrate the superiority and validity of the proposed optimal flocking algorithm compared with existing approaches.

Index Terms—Distributed safety barrier certificate, high-order control barrier function, optimal flocking algorithm, selective relaxation variable

I. INTRODUCTION

FOR the multi-agent system (MAS), distributed control has been studied in various fields thanks to its clear advantages over the centralized approach in terms of efficiency, robustness, flexibility, scalability, and reliability [1]. Many studies for MAS have designed controllers using control Lyapunov function theory to accomplish given missions, e.g., formation [2], flocking [3], containment control [4], and target tracking [5]. However, these MAS controllers have limitations

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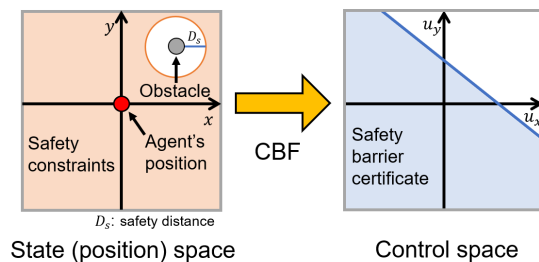


Fig. 1: Mapping of safety constraints by the CBF.

in that their control performance relies on parameter tuning and cannot explicitly guarantee collision avoidance among agents.

To address this issue, the optimal control problem has been studied to achieve the control objective optimally and ensure collision avoidance. The optimal control problem is formulated as an optimization problem incorporating an objective function and safety constraints according to the control objective and system requirements. At every control instant, the optimal control input is generated by solving the given optimization problem. For instance, the model predictive control (MPC) method generates an optimal control input within safety constraints (e.g., collision avoidance) by computing the future state in a receding horizon step [6], [7]. However, since the computational complexity of the MPC method increases with the horizon step, the use of the control barrier function (CBF) in optimal control has gained more attention recently [8]. According to the recent formulation of (zeroing) CBF in [9], the CBF yields safety barrier certificate (SBC) that maps safety constraints expressed in the system state space to an admissible set in the control space as illustrated in Fig. 1. Since the control input generated within the SBC ensures that states remain within safety constraints without explicitly calculating the future states, it enables the generation of a safe optimal control input in real-time. Thanks to this property, the CBF is utilized in various fields, such as adaptive cruise control [10], crossing multi-lane roads [11], and robotic systems [12]. Nevertheless, these studies are restricted in that the relative degree of the CBF is one. To handle high relative degree constraints, the exponential CBF is proposed based on the pole placement control technique in linear control theory [13].

High-order CBF (HOCBF) [14], [15] has also been proposed as a more general CBF for high relative degree where the system of CBF is not limited to linear compared with the exponential CBF.

In MAS, the distributed safety barrier certificate (DSBC) is designed by distributing the centralized safety barrier certificate (CSBC) to each agent for distributed implementation of the optimal control problem [16]. Intuitively, each agent should consider all neighboring agents' states to ensure collision-free among agents, and thus the admissible control space is the intersection of the DSBC for all neighbors. However, with a naive DSBC implementation, such as equal distribution of the CSBC, a feasibility problem could occur wherein the admissible control space becomes empty (i.e., there is no intersection of all DSBCs) [16]–[18]. Several approaches have been proposed to deal with the feasibility problem. One of the approaches is to expand the DSBC by introducing relaxation variables [19]. The relaxation method works by increasing (or optimizing) the relaxation variables to expand the DSBC when the admissible control space is empty. In addition, to address the problem of no feasible control input for collision avoidance, Li *et al.* designed a CBF considering the maximum braking force [16]. The CBF is designed to maintain a safety distance until both agents stop by applying the maximum braking force. Xiao *et al.* proposed sufficient conditions that ensure the feasibility of safety constraints and control bounds [20]. Here, the sufficient conditions require that the hyperplane and direction of Lie derivatives of safety constraints and candidate feasibility constraints are identical.

Nevertheless, the aforementioned methods have several limitations. First, the relaxation method increases the dimensionality of the optimization variable in proportion to the number of neighbors, which increases computational complexity and thus making it non-scalable for the MAS. Second, considering that the designed CBF in [16] assumes that both agents use the maximum braking force to maintain a safety distance, collision avoidance is guaranteed only when the maximum braking force is applied to both agents simultaneously. However, collision avoidance might not be guaranteed by the maximum braking method presented in [16] in a distributed setting. This is because, in situations where only one agent encounters a feasibility problem and has to use maximum braking, other agents can still find feasible control inputs within their admissible control space. As a result, these agents may continue to move and potentially cause a collision with the agent using the maximum braking method. Lastly, finding the candidate feasibility constraints satisfying the sufficient conditions described in [20] might be difficult. Moreover, further research is needed to address feasibility constraints for multiple safety constraints across all neighbors in the MAS.

In this study, we formulate an optimal control problem that satisfies safety constraints by using the concept of HOCBF to achieve the safe flocking of the MAS. In particular, this study aims to effectively enhance feasibility to alleviate the feasibility problem frequently occurring in flocking control of the MAS. To this end, first, we propose a DSBC to be assigned differently according to the collision risk of each agent related to safety constraints unlike the existing equal dis-

tribution approach [16]–[18]. The proposed method expands the DSBC for agents with a high collision risk while shrinking the DSBC for agents with a low collision risk. Second, we propose a selective relaxation method with less increase in computation by selectively applying the relaxation variables. The proposed method adds the relaxation variables only when the admissible control space is small or empty compared to the existing relaxation method adding the relaxation variables to all safety constraints. Specifically, an arbitrary element within the control bound is chosen first such as a control input that maintains the current velocity (i.e., zero acceleration command), and the relaxation variables are included in the optimization process only if the DSBC does not contain this element. This effectively solves the distributed optimization problem with the fewer optimization variables by applying the relaxation method only to some of the neighboring agents which cause the feasibility problem, instead of applying the relaxation method to all neighboring agents in the existing work [16], [19]. Through comprehensive numerical simulations, the superiority and effectiveness of the proposed safety-critical optimal flocking control algorithms are demonstrated.

The main contributions of this paper are as follows:

- 1) Efficient distributed flocking control: distributed optimal flocking control is proposed with the HOCBF, which guarantees a high relative degree of safety constraints. The safe optimal control input generated by the HOCBF can achieve flocking efficiently while avoiding collisions among agents, provided that the admissible control space is not empty;
- 2) Enhance feasibility in the optimal control problem of the MAS: the DSBC is assigned according to the relative collision risk of each agent for feasibility enhancement of the distributed optimal control problem; and
- 3) Reduce computation complexity: the relaxation variable is selectively applied to the DSBC only when the admissible control space is small or empty to reduce computational complexity.

The rest of this paper is organized as follows. In Section II, we present the preliminaries related to the CBF. Section III presents the problem formulation for optimal flocking control. For feasibility enhancement, the DSBC considering a collision risk and the selective relaxation method are proposed in Sections IV and V, respectively. In Section VI, numerical simulations demonstrate the performance of the proposed feasibility enhancement methods in optimal flocking control. Finally, concluding remarks and areas of future study are provided in Section VII.

II. PRELIMINARIES

This study focuses on the behavior of mobile MAS applications where the agent's dynamic model is the double integrator. For the sake of generality, consider control affine systems in the form of

$$\dot{x} = f(x) + g(x)u, \quad (1)$$

where the state $x \in \mathbb{R}^n$, control input $u \in U \subset \mathbb{R}^r$, U is the control bound, and $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times r}$ are locally Lipschitz. In mobile MAS operations, ensuring safety

(i.e., collision avoidance) is essential. Let $C \subset \mathbb{R}^n$ be the safe set where we want the agent states to stay, defined as

$$C := \{x \in \mathbb{R}^n \mid h(x) \geq 0\},$$

where $h : \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuous function and $h(x) \geq 0$ is a safe condition for the system (1). For the agent states to stay in the safe set C , C must be forward invariant which is defined as follows:

Definition 1 (Forward Invariant): The set C is said to be forward invariant with respect to system (1) if $x(t) \in C \forall t \geq 0$ for every $x(0) \in C$.

The purpose of CBFs is to ensure the forward invariance of the safe set C . Before defining CBF formally, the definitions of the extended class $\mathcal{K}$ function and time derivative of the function h are described. The time derivative of the function h can be expressed as

$$\frac{dh(x)}{dt} = \frac{\partial h(x)}{\partial x} \dot{x} = \frac{\partial h(x)}{\partial x} (f(x) + g(x)u),$$

which can also be formulated by Lie derivative as

$$\frac{dh(x)}{dt} = L_f h(x) + L_g h(x)u.$$

Definition 2 (Extended Class $\mathcal{K}$ Function [9]): A continuous function $\alpha : (-b, a) \rightarrow \mathbb{R}$ for some $a, b > 0$ is called an extended class $\mathcal{K}$ function if it is strictly increasing and $\alpha(0) = 0$.

The following definition of CBF can be regarded as zeroing CBF introduced in [9]:

Definition 3 (Control Barrier Function [9]): Given the dynamical system (1) and a set $C \subset \mathbb{R}^n$ for a smooth function $h : \mathbb{R}^n \rightarrow \mathbb{R}$, the function h is called a control barrier function (CBF), if there exists an extended class $\mathcal{K}$ function α such that

$$\sup_{u \in U} \{L_f h(x) + L_g h(x)u + \alpha(h(x))\} \geq 0.$$

According to the definition of the CBF, the admissible control space $U^{CBF}(x)$ for a given CBF h can be defined as

$$U^{CBF}(x) := \{u \in U \mid L_f h(x) + L_g h(x)u + \alpha(h(x)) \geq 0\}.$$

Now, we can guarantee forward invariance for the safe set C of the system (1) through the following Lemma:

Lemma 1 [9]: Given a set $C \subset \mathbb{R}^n$ and a CBF h , any Lipschitz continuous controller $u \in U^{CBF}(x)$ at each time step for the system (1) renders the set C forward invariant.

Next, we present the definition of relative degree for dynamic systems and introduce high-order CBF (HOCBF), which is a generalized version of the CBF to deal with a high relative degree.

Definition 4 (Relative Degree): The relative degree of an output y with respect to the system (1) is the number of times it needs to be differentiated along its dynamics until the control input u is explicitly shown in the time derivative.

The HOCBF is designed for a high relative degree of the function h . For the system (1), a sequence of functions is defined based on the function h with relative degree m as

$$\begin{aligned} \psi_0(x) &:= h(x), \\ \psi_1(x) &:= \dot{\psi}_0(x) + \alpha_1(\psi_0(x)), \\ &\vdots \end{aligned} \tag{2}$$

$$\psi_{m-1}(x) := \dot{\psi}_{m-2}(x) + \alpha_{m-1}(\psi_{m-2}(x)),$$

where $\psi_i : \mathbb{R}^n \rightarrow \mathbb{R}, i \in \{0, \dots, m-1\}$ and $\alpha_i : \mathbb{R} \rightarrow \mathbb{R}, i \in \{1, \dots, m-1\}$ is a $(m-i)^{th}$ order differentiable extended class $\mathcal{K}$ function. Associated with (2), a further sequence of safe sets $C_i, i \in \{0, \dots, m-1\}$ are defined as

$$\begin{aligned} C_0 &:= \{x \in \mathbb{R}^n \mid \psi_0(x) \geq 0\}, \\ C_1 &:= \{x \in \mathbb{R}^n \mid \psi_1(x) \geq 0\}, \\ &\vdots \\ C_{m-1} &:= \{x \in \mathbb{R}^n \mid \psi_{m-1}(x) \geq 0\}. \end{aligned} \tag{3}$$

Definition 5 (High-Order Control Barrier Function [14]):

Given the dynamical system (1) and sets $C_i \subset \mathbb{R}^n, i \in \{0, \dots, m-1\}$ defined by (3) for a smooth function $h : \mathbb{R}^n \rightarrow \mathbb{R}$, the function h is called a high-order control barrier function (HOCBF) of relative degree m , if there exist $(m-i)^{th}$ order differentiable extended class $\mathcal{K}$ functions $\alpha_i : \mathbb{R} \rightarrow \mathbb{R}, i \in \{1, \dots, m-1\}$ and an extended class $\mathcal{K}$ function $\alpha_m : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\begin{aligned} \sup_{u \in U} \{L_f^m h(x) + L_g L_f^{m-1} h(x)u + \sum_{i=1}^{m-1} \alpha_i^{(m-i)}(\psi_{i-1}(x)) \\ + \alpha_m(\psi_{m-1}(x))\} \geq 0, \end{aligned}$$

where L_f^m denotes Lie derivatives along f m times and $\alpha_i^{(m-i)}$ denotes the $(m-i)^{th}$ time derivative function.

Note that equation in the definition of HOCBF is the same as $\sup_{u \in U} \{\dot{\psi}_{m-1}(x) + \alpha_m(\psi_{m-1}(x)) \geq 0\}$, which can be expressed with $h, \alpha_i, i \in \{1, \dots, m\}$, and its derivatives. The HOCBF is a general form of the CBF in which the HOCBF with the relative degree of one reduces to the common CBF form. According to the definition of the HOCBF, the admissible control space $U^{HOCBF}(x)$ for a given HOCBF h can be defined as

$$U^{HOCBF}(x) := \{u \in U \mid \dot{\psi}_{m-1}(x) + \alpha_m(\psi_{m-1}(x)) \geq 0\}. \tag{4}$$

Lemma 2 [14]: Given a set $\bar{C} \subset \mathbb{R}^n$ defined by $\bar{C} = \bigcap_{i=0}^{m-1} C_i$ and a HOCBF h , any Lipschitz continuous controller $u \in U^{HOCBF}(x)$ for the system (1) renders the set $\bar{C}$ forward invariant.

III. PROBLEM FORMULATION FOR SAFETY-CRITICAL OPTIMAL FLOCKING CONTROL

This study deals with designing a controller which enables multiple agents to achieve stable flocking states from an arbitrary initial state while ensuring safety. Flocking refers to a self-organized collective motion of multiple agents such as a loose formation of birds in flight. The flocking behavior is composed of three elements: cohesion, alignment, and separation. Cohesion represents the group concentration, in which agents distant from the center of the group would move

inward to the center to maintain a loose formation. Alignment enables the collective motion of the group by synchronizing the individual agent's velocity with adjacent agents. Separation functions to avoid collisions between too close to each other. To achieve a stable and safe flocking behavior, we propose a framework for the safe optimal flocking control. This framework enables optimal controller design in which the required time and consumed energy is minimized to reach stable flocking states, i.e., position and velocity deviations from the average position and velocity of all agents are unchanged. While achieving flocking states, collision avoidance is dealt with the safety constraints expressed as the HOCBF to keep a certain safety distance among agents.

A. System Dynamics

Similar to [14], [16], we consider a double integrator model for the individual agent where the MAS consists of N agents. The state of each agent is composed of the position and velocity, and the dynamics of agent i can be written as

$$\dot{x}_i = Ax_i + Bu_i, \quad (5)$$

where $i \in M$, $M = \{1, \dots, N\}$, $x_i = [p_i^T v_i^T]^T \in \mathbb{R}^{2n}$, $p_i \in \mathbb{R}^n$ is the position vector, $v_i \in \mathbb{R}^n$ is the velocity vector, and $u_i \in \mathbb{R}^n$ is the control input vector. The state transition matrix A and control input matrix B have the following form:

$$A = \begin{bmatrix} 0_n & I_n \\ 0_n & 0_n \end{bmatrix}, B = \begin{bmatrix} 0_n \\ I_n \end{bmatrix},$$

where I_n and 0_n are the n -dimensional identity matrix and zero matrix, respectively. Due to physical limitations, the magnitude of the velocity is bounded by $\|v_i\| \leq v_{max}$. Similarly, the control bound set can be defined as

$$U_i = \{u_i \in \mathbb{R}^n \mid \|u_i\| \leq u_{max}\}. \quad (6)$$

Note that the MAS considered in this study consists of homogeneous agents, representing that all agents have common physical properties.

B. Objective Function

The objective function to reach the stable flocking states is designed to effectively achieve cohesion, alignment, and desired velocity with minimum control efforts. Besides, considering that the computational complexity of the centralized approach to solve optimization problems for the MAS is large, the objective function is designed in a distributed way so that each agent can derive an optimal solution using only its own and neighbor's information. According to these requirements, the objective function for agent i can be formulated as

$$\begin{aligned} J_i(u_i(T)) = & \int_t^{t+\delta t} \sum_{j \in N_i} \left(\|v_i(\tau) - v_j^i(\tau)\|^2 \right. \\ & + \lambda_1 (\|p_i(\tau) - p_j^i(\tau)\| - R_d)^2 \\ & \left. + \lambda_2 \|u_i(\tau)\|^2 + \lambda_3 \|v_i(\tau) - v_d\|^2 \right) d\tau, \end{aligned}$$

where $T \in [t, t + \delta t]$, λ_1, λ_2 and $\lambda_3 > 0$ are weights, R_d is the desired distance between neighbor agents, v_d is the desired

velocity, and p_j^i and v_j^i are predicted state of the neighboring agent with respect to agent i . The control input $u_i(\tau)$ is generated such that the objective function is minimized in time interval $\tau \in [t, t + \delta t)$. The set of neighbors N_i is defined as

$$N_i = \{j \in M \setminus \{i\} \mid \|p_{ij}\| < R_c\}$$

where $p_{ij} = p_i - p_j$ is the relative position between agents and R_c is the communication range. It is worthwhile noting that the communication topology is time-varying because the neighborhood set is defined based on relative distance. Four terms in the objective function are related to the alignment, cohesion, minimizing the control input, and achieving the desired velocity, respectively. Since it is difficult to accurately predict the future state of the neighbors with only the information available to agent i , the future state of neighbors is predicted based on the assumption that $u_j = 0$ as done in [6], [21]. Defining the objective function over a short time interval minimizes uncertainties related to changes in time-varying communication topology and predicted neighbors' future states, while also reducing the time required to compute the optimal control input. This approach is effective because CBF can ensure safety without needing to predict states far into the future.

In practice, most controllers are implemented digitally, resulting in point-wise generation of controllers in discrete time using zero-order hold. In the discretization process, the safety constraints should be maintained throughout the interval by accounting for the inter-sampling effect [22]. The objective function defined in continuous time could be reconstructed in discrete time as in [20] to generate the discrete optimal control input as

$$\begin{aligned} J_i(u_i(k)) = & \sum_{j \in N_i} \left(\|v_i(k+1) - v_j(k)\|^2 \right. \\ & + \lambda_1 (\|p_i(k+2) - (p_j(k) + 2\Delta t v_j(k))\| - R_d)^2 \\ & \left. + \lambda_2 \|u_i(k)\|^2 + \lambda_3 \|v_i(k+1) - v_d\|^2 \right), \end{aligned} \quad (7)$$

where k is the discrete time index, and Δt is the sampling time of discretization. Notation of $v_i(k+1)$ is the updated velocity in discrete time by discretization of the system dynamics (5) where $p_i(k+1) = p_i(k) + \Delta t v_i(k)$ and $v_i(k+1) = v_i(k) + \Delta t u_i(k)$. The future state of neighbors is updated based on the current state assuming $u_j(k) = 0$ which brings $v_j(k+1) = v_j(k)$ from the point of view of agent i . Similarly, at $k+2$ instant where the control input $u_i(k)$ affects the position, the control input is generated to keep its position at the desired distance of R_d with respect to its neighbors. The neighbor position is predicted as $p_j(k+2) = p_j(k) + 2\Delta t v_j(k)$ from the point of view of agent i . The optimal control input obtained by minimizing the objective function (7) is applied at the current time step and held constant for the interval $[k, k+1)$. The state is updated using the system dynamics (5), and the entire procedure is repeated.

C. Safety Constraints

In this subsection, collision avoidance related to separation between agents for flocking is formulated as the constraint to

be used in the optimal control problem. The safety constraints, represented by the system states to avoid collision between agents, are represented as

$$h(x_i, x_j) = \frac{1}{2}(\|p_{ij}\|^2 - D_s^2) \geq 0, \quad (8)$$

where D_s is a safety distance. It is worthwhile pointing out that the relative degree of (8) is two. Hereafter, $h(x_i, x_j)$ is denoted as h_{ij} for notation simplicity. In order to satisfy the safety constraints expressed as the system states, one option is to use the MPC calculating an unsafe set through the future state in a receding horizon step; however, it requires large computation. On the other hand, mapping the constraint of the state space to the control space through the HOCBF has the advantage of efficiently deriving the optimal control input without additional computation for the constraints.

In the next section, the concept of HOCBF is applied to safety constraints (8). The centralized safety barrier certificate (CSBC) of the safety constraints is first derived, yielding constraints in the control space of agents i and j . Then, to solve the optimal control problem in a distributed way, the CSBC is divided into the distributed safety barrier certificates (DSBCs) which are distributed into agents i and j . However, the intersection of individually-assigned DSBCs within the control bound set in (6) might become empty in challenging conditions, resulting in the feasibility problem, i.e., no optimal control input while satisfying the forward invariance of the safe set exists. To alleviate this issue, we propose a method of distributing DSBC effectively considering the collision risk of each agent, which is introduced in detail in Section 4.

IV. SAFETY BARRIER CERTIFICATE

A. Centralized Safety Barrier Certificate

Since the safety constraints (8) is coupled with the neighbor agent's state, it is natural to introduce the control space that guarantees collision avoidance in a centralized way while treating h_{ij} in (8) as the HOCBF. Considering that the relative degree of h_{ij} is two, safe sets and admissible control space are defined by using (3) and (4), respectively, as

$$\begin{aligned} C_0 &= \{\mathbf{x} \in \mathbb{R}^{2nN} | \psi_{0,ij} \geq 0\}, \\ C_1 &= \{\mathbf{x} \in \mathbb{R}^{2nN} | \psi_{1,ij} \geq 0\}, \\ U^{HOCBF}(\mathbf{x}) &= \{\mathbf{u} \in \mathbb{R}^{Nn} | \dot{\psi}_{1,ij} + \alpha_2(\psi_{1,ij}) \geq 0\}, \end{aligned} \quad (9)$$

where $\forall i \in M, \forall j \in N_i$, and $\mathbf{x} = [x_1^T, \dots, x_N^T]^T$ and $\mathbf{u} = [u_1^T, \dots, u_N^T]^T$ are the augmented state and control input of all N agents, respectively. $\psi_{0,ij} = \psi_0(x_i, x_j)$ and $\psi_{1,ij} = \psi_1(x_i, x_j)$ are the simplified notation of ψ functions. With the use of the extended class $\mathcal{K}$ function as a linear function $\alpha_i(x) = \gamma_i x$ where $\gamma_i > 0$, functions ψ are defined by (2) as $\psi_{0,ij} = h_{ij}$ and $\psi_{1,ij} = \dot{\psi}_{0,ij} + \alpha_1(\psi_{0,ij}) = p_{ij} \cdot v_{ij} + \frac{\gamma_1}{2}(\|p_{ij}\|^2 - D_s^2)$ where $v_{ij} = v_i - v_j$ is the relative velocity and $\cdot$ is the inner product notation. The left side of the inequality in (9) is then expressed by system states as

$$\begin{aligned} \dot{\psi}_{1,ij} + \alpha_2(\psi_{1,ij}) &= (p_{ij} \cdot u_{ij} + \|v_{ij}\|^2 + \gamma_1 p_{ij} \cdot v_{ij}) \\ &\quad + \gamma_2(p_{ij} \cdot v_{ij} + \frac{\gamma_1}{2}(\|p_{ij}\|^2 - D_s^2)), \end{aligned} \quad (10)$$

where $u_{ij} = u_i - u_j$ is the relative control input. The CSBC appears by combining (10) with the inequality in (9) as

$$-p_{ij} \cdot u_{ij} \leq a_{ij} + b_{ij} + c_{ij}, \quad (11)$$

where $a_{ij} = \|v_{ij}\|^2 + \frac{\gamma_1 \gamma_2}{2} \|p_{ij}\|^2 \geq 0$, $b_{ij} = -\frac{\gamma_1 \gamma_2}{2} D_s^2 \leq 0$, and $c_{ij} = (\gamma_1 + \gamma_2) p_{ij} \cdot v_{ij}$.

Definition 6 (safety): The safety of the MAS with dynamics of each agent given in (5) is achieved if the augmented state $\mathbf{x}(t) \in \bar{C} (= C_0 \cap C_1), \forall t \geq 0$.

With the definition of safety and the admissible control space defined by the CSBC (11), the following theorem is presented to guarantee the safety of the MAS in a centralized way.

Theorem 1: Given the dynamics (5) of each agent for the MAS, if $\mathbf{x}(0) \in \bar{C}$ and the controller $\mathbf{u}(t)$ is generated within the admissible control space (9) and the control bound formulated by (6), then the safety of the MAS is guaranteed.

Proof: As ensured by the HOCBF in Lemma 2, the controller $\mathbf{u}(t)$ makes $\bar{C}$ being forward invariant if $\mathbf{x}(0) \in \bar{C}$. Therefore, since $\mathbf{x}(0) \in \bar{C}$, $\mathbf{x}(t)$ will stay in $\bar{C}$ for all time, which means the safety of the MAS is guaranteed. ■

Generally, the minimally invasive approach is used to generate the optimal control input, ensuring safety by Theorem 1, when a nominal controller is available [19, 20]. The goal of the minimally invasive controller is to make the optimal control input closely follow the nominal controller. This is achieved by constructing an objective function with a quadratic cost that minimizes the difference between the nominal $\hat{\mathbf{u}}$ and the control input $\mathbf{u}$ as follows:

$$\begin{aligned} \mathbf{u}^* &= \underset{\mathbf{u} \in \mathbb{R}^{Nn}}{\operatorname{argmin}} \|\mathbf{u} - \hat{\mathbf{u}}\|^2, \\ \text{s.t. } \mathbf{u} &\in U^{HOCBF}(\mathbf{x}), \\ u_i &\in U_i, \forall i \in M. \end{aligned}$$

B. Distributed Safety Barrier Certificate Considering Collision Risk

A central coordination unit in the MAS can limit scalability and robustness, as it may become overloaded with data as the number of agents grows, causing delays and creating a single point of failure. Instead, distributed control allows decision-making to be spread across agents, reducing reliance on the coordination unit and thus enhancing system performance. Therefore, it is often more effective for each agent to solve optimal control problems in a distributed manner by distributing the CSBC to individual agents.

The DSBC is a form in which the CSBC is partitioned into smaller subsets while safety can still be guaranteed by generating the individual control input within its assigned DSBC. Before describing the distribution technique for the DSBC, the admissible control space for agent i in a distributed way can be defined as

$$U_i^{HOCBF}(\mathbf{x}_{N_i}) = \{u_i \in \mathbb{R}^n | -p_{ij} \cdot u_i \leq f_{ij}\}, \quad (12)$$

where $j \in N_i$, $\mathbf{x}_{N_i} = [x_i^T, x_{j_1}^T, \dots, x_{j_{|N_i|-1}}^T]^T$ is the state of its own and all neighbors, and f_{ij} is the assigned value for the DSBC which satisfies $f_{ij} + f_{ji} = a_{ij} + b_{ij} + c_{ij}$ in (11). Note

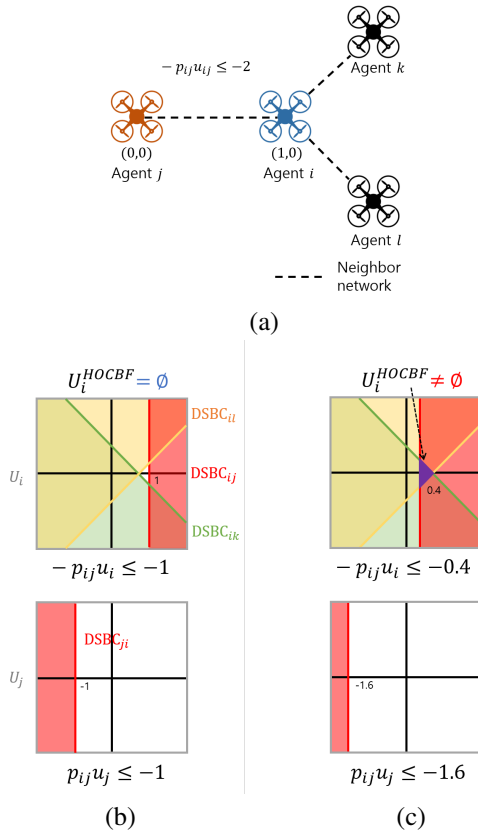


Fig. 2: Feasibility enhancement by considering a collision risk. (a) Agents' position. (b) Infeasible case. (c) Feasible case.

that the control input in the DSBC is changed to u_i instead of u_{ij} in the CSBC. As one way of assigning the DSBC for homogeneous agents in the literature [16]–[18], f_{ij} and f_{ji} can be made the same for each agent as

$$\begin{aligned} -p_{ij} \cdot u_i &\leq f_{ij}, \\ p_{ij} \cdot u_j &\leq f_{ji}, \\ f_{ij} &= \frac{1}{2}a_{ij} + \frac{1}{2}b_{ij} + \frac{1}{2}c_{ij}, \end{aligned} \quad (13)$$

where $f_{ij} = f_{ji}$ represents the equal distribution of the DSBC.

However, equal distribution does not take into account the different collision risks of agents. Since the admissible space of an agent with a high collision risk can be easily empty, this can cause the feasibility problem. Let us consider an illustrated example in Fig. 2(a), where agent i confronts a high collision risk and focus on the DSBC between agents i and j . More specifically, the positions of agents i and j in the two-dimensional environment are $(1,0)$ and $(0,0)$, respectively, and the CSBC is set to $-p_{ij} \cdot u_{ij} \leq -2$. Figure 2(b) shows the equal distribution case where the intersection of all DSBC lines with three neighbors of agent i (i.e., the admissible control space of agent i) becomes empty as agent i is constrained not only by agent i but also agents k and l . Meanwhile, the admissible control space of agent j is large enough. This feasibility problem could be solved by changing the DSBC distribution. For instance, the right-hand side of the DSBC of agent i could be increased from -1 to -0.4 , and that of the

DSBC of agent j could be decreased from -1 to -1.6 , as shown in Fig. 2(c). Then, there exists the admissible control space of agent i , indicated as the purple-colored area in Fig. 2(c).

Inspired by this, we propose a distribution method of the DSBC to enhance feasibility. The DSBC coefficients are determined by the HOCBF which is directly related to the collision risk of agents. The proposed method expands the DSBC of an agent with a relatively high collision risk while shrinking the DSBC of the pairwise agent with a relatively low collision risk. Here, expanding the DSBC means enlarging a control space region, which can be achieved by increasing f_{ij} in (12). In this study, compared with (13), f_{ij} is increased by increasing $a_{ij} (\geq 0)$ and decreasing $b_{ij} (\leq 0)$ based on the CSBC in (11), which is represented for agent i as

$$f_{ij} = (1 - \eta_{ij})a_{ij} + \eta_{ij}b_{ij} + \frac{1}{2}c_{ij}, \quad (14)$$

$$\eta_{ij} = \frac{\bar{h}_i}{\bar{h}_i + \bar{h}_j}, \quad (15)$$

where $\bar{h}_i = \frac{1}{|N_i|} \sum_{k \in N_i} h_{ik}$, η_{ij} is a collision risk distribution coefficients satisfying $0 \leq \eta_{ij} \leq 1$, and $\eta_{ij} + \eta_{ji} = 1$. Note that to calculate η_{ij} in (15), agent i only needs a single scalar value $\bar{h}_j$ from its neighboring agent j . This means that the calculation can be performed by exchanging just this one scalar value, without needing to know the neighbor sets of its neighbors, i.e., N_j . In (15), η_{ij} is used to compare the average HOCBF between neighboring agents i and j . A lower average HOCBF indicates a higher collision risk. If η_{ij} is less than 0.5, it means that the average HOCBF of agent i is lower than that of agent j , indicating a relatively high collision risk. Therefore, considering η_{ij} represents the relative collision risk, the distribution coefficients of f_{ij} in (14) expand the DSBC of the high collision risk agent having a small η_{ij} . Note that if the relative collision risk of agents i and j are the same, the collision risk distribution is the same as the equal distribution in (13).

Considering inappropriate neighbors can affect the calculation of the collision risk distribution coefficient. The admissible control space is formulated by the intersection of the SBCs of all neighbors within the communication range. If the number of neighbors increases, the associated computation of the SBCs also increases. Furthermore, the proposed method for determining the distribution coefficient of the DSBC in (15) may be inappropriate. In Fig. 3, considering that the collision risk increases as h decreases, agent i has a higher collision risk with agent k than agent j . On the other hand, although agents i and l are neighbors, they are far enough apart that there is no collision risk. In this situation, η_{ij} in (15) is calculated as $990/(990+10)$, which is larger than 0.5, and thus, the collision risk of agent i can be regarded as lower than that of agent j ; it is contrary to the actual situation. This issue arises because agent i considers distant agent l . To address this issue, the neighbors should be selected considering a potential collision risk as well as a communication range R_c ; this concept is termed as selecting reduced neighbors. In Fig. 3, with reduced neighbors, η_{ij} is $5/(5+10)$, which is less than 0.5, representing

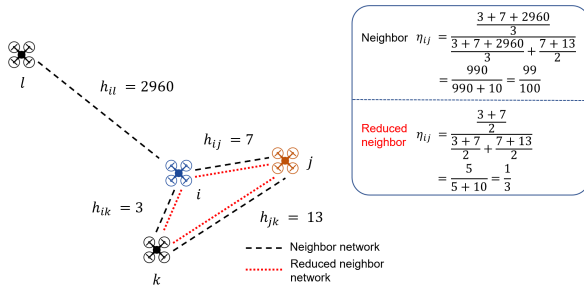


Fig. 3: Illustration of neighbor and reduced neighbor network.

a higher collision risk of agent i than agent j .

Formally, agents that satisfy the CSBC (11) (i.e., safety is guaranteed) should not be included in the reduced neighbor set. The criterion for reduced neighbors with a high collision risk (i.e., not satisfying the CSBC condition) is expressed with a certain relative distance R_{re} as in the below theorem.

Theorem 2: To guarantee the safety of the MAS, agent i only needs to form the HOCBF with its reduced neighbors N_i^r , defined as

$$N_i^r = \{j \in N_i \mid \|p_{ij}\| < R_{re}\},$$

$$R_{re} = y + \sqrt{y^2 + D_s^2 - \frac{8}{\gamma_1 \gamma_2} v_{max}^2}, \quad (16)$$

where R_{re} is the distance of the reduced neighbors and $y = \frac{2}{\gamma_1 \gamma_2} (u_{max} + (\gamma_1 + \gamma_2) v_{max})$.

Proof: To figure out the condition satisfying (11) regardless of the current states and control action, we should consider the worst-case scenario where two agents move across to each other at the maximum speed and acceleration. In the worst-case, the relative velocity and the control input are $v_{ij} = -2v_{max}p_{ij}/\|p_{ij}\|$ and $u_{ij} = -2u_{max}p_{ij}/\|p_{ij}\|$, respectively. Substituting them into (11) gives

$$g(\|p_{ij}\|) = \frac{\gamma_1 \gamma_2}{2} \|p_{ij}\|^2 - 2(u_{max} + (\gamma_1 + \gamma_2) v_{max}) \|p_{ij}\| + 4v_{max}^2 - \frac{\gamma_1 \gamma_2}{2} D_s^2 \geq 0. \quad (17)$$

Equation (17) is expressed as a quadratic function for the relative distance $\|p_{ij}\|$, and the solution of this for $\|p_{ij}\|$ is

$$\|p_{ij}\| \geq \frac{2}{\gamma_1 \gamma_2} (u_{max} + (\gamma_1 + \gamma_2) v_{max}) + \sqrt{\left(\frac{2}{\gamma_1 \gamma_2} (u_{max} + (\gamma_1 + \gamma_2) v_{max})\right)^2 + D_s^2 - \frac{8}{\gamma_1 \gamma_2} v_{max}^2} = R_{re}.$$

where the details for this computation can be found in Appendix. The above equation means that if the distance between the two agents is larger than R_{re} , (11) is always satisfied regardless of the agents' state and the control action. Therefore, for the safety of MAS, agent i only needs to consider the HOCBF with the agents belonging to N_i^r . ■ With the collision risk distribution of DSBC and the reduced neighbors, the following theorem is presented to guarantee the safety of the MAS in a distributed way.

Theorem 3: Given the dynamics (5) of each agent for the MAS, if $\mathbf{x}(0) \in \bar{C}$ and the controller $u_i(t)$ for all agent $i \in M$ is generated within the admissible control space (12) with reduced neighbors (16) and the modified f_{ij} (14) and the control bound (6), then the safety of the MAS is guaranteed.

Proof: If each agent controller $u_i(t)$ belongs to (12), then $\mathbf{u}(t)$ belongs to (9) for all agent $j \in N_i^r$. Besides, due to Theorem 2, $\mathbf{u}(t)$ still belongs to (9) for all agent $j \in N_i \setminus N_i^r$. Therefore, by Theorem 1, the safety of MAS is guaranteed. ■

Remark 1: Equation (7) still aims to minimize the objective function with the neighbor N_i . Besides, the reduced neighbors N_i^r become the same as neighbors N_i when the distance R_{re} of the reduced neighbors is larger than the communication range R_c .

Based on the proposed collision risk distribution of DSBC, the optimal flocking control problem for agent i is formulated as

$$u_i^* = \underset{u_i \in \mathbb{R}^n}{\operatorname{argmin}} J_i,$$

$$\text{s.t. } -p_{ij} \cdot u_i \leq f_{ij},$$

$$\|u_i\| \leq u_{max},$$

where f_{ij} is from (14) with η_{ij} while considering the reduced neighbors N_i^r . Note that the methods proposed in this study to enhance the feasibility of CBF are not limited to safety constraints. Indeed, they can be applied to any kind of inequality constraints, such as control bounds or feasibility constraints [20]. These methods are particularly effective in environments where multiple agents need to interact each other to satisfy constraints. However, the proposed method can still result in infeasibility. Further research is required to guarantee both safety and feasibility.

V. RELAXED SAFETY BARRIER CERTIFICATE

In this section, we introduce another feasibility enhancement method, which is the use of relaxation variables to solve the feasibility problem when the admissible control space is empty. However, since the relaxation variables are included in the optimization problem as additional optimization variables, the computational complexity increases significantly. To mitigate this, we propose a method of selectively applying the relaxation variable. It is worthwhile noting that, in contrast to the approach presented in Section IV where the individual DSBC is expanded or reduced without any modification to the CSBC, the relaxation method expands the CSBC.

A. Modified Relaxation Method

By introducing the relaxation variable, the relaxation method can expand the admissible control space, and it has been applied in several studies. For instance, the relaxation method was employed in [19] to ensure feasibility and in [16] to enlarge the admissible control space due to the conservative collision avoidance strategy of the DSBC relative to the CSBC.

The method of relaxation can be employed by modifying the inequality of (9) to $\dot{\psi}_{1,ij} + \omega_{ij} \alpha_2(\psi_{1,ij}) \geq 0$, where $\omega_{ij} \in [1, \infty)$ is a continuous relaxation variable [16]. However, this conventional relaxation method (CRM) could not expand

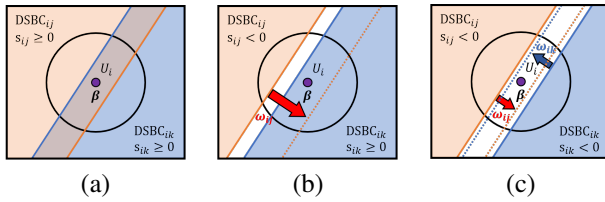


Fig. 4: Classifying the cases in which selective relaxation methods are applied. (a) Both DSBCs contain β . (b) One DSBC contains β . (c) Neither of DSBCs contain β .

the admissible control space when $\psi_{1,ij}$ is negative. To address this issue, a relaxation variable should be applied to the positive term of the SBC to expand the admissible control space regardless of the value of $\psi_{1,ij}$. This is termed as a modified relaxation method (MRM) in this study, which can be expressed by modifying the inequality of (11) as

$$-p_{ij} \cdot u_{ij} \leq w_{ij}a_{ij} + b_{ij} + c_{ij},$$

where a_{ij} is the positive term of SBC and w_{ij} is the centrally computed relaxation variable. This equation shows that the CSBC can be expanded by the relaxation variable w_{ij} when the admissible control space is small or empty regardless of the value of $\psi_{1,ij}$. Note that this equation is equivalent to (11) when w_{ij} is one. Furthermore, the proposed relaxation method can be expressed using the relaxation method in [16], thereby satisfying the relaxation condition outlined in [16]. Similarly, the relaxed DSBC for agent i with the collision risk distribution coefficients proposed in Section IV can be rewritten as

$$-p_{ij} \cdot u_i \leq (1 - \eta_{ij})\omega_{ij}a_{ij} + \eta_{ij}b_{ij} + \frac{1}{2}c_{ij}, \quad (18)$$

where ω_{ij} is the relaxation variable of agent i for agent j in a distributed setting. Note that in a centralized manner, $w_{ij} = w_{ji}$, but in a distributed manner, w_{ij} can be different from w_{ji} .

B. Selective Relaxation Method

Introducing the relaxation variable to expand the admissible control space allows for greater adaptability to the situation, thereby improving feasibility. However, the computational burden increases substantially as the dimensionality of the optimization variable increases from the control input dimension n to $n + |N_i^s|$ due to the need to account for the relaxation variables, as shown in the previous study [16]. Furthermore, when control bound (6) is already a part of the DSBC (i.e., no feasibility issue), there is no need to use the relaxation method. This implies that the relaxation variables should be selectively applied according to the DSBC condition in order to reduce computational burden.

In this study, to effectively enhance feasibility with fewer relaxation variables, we first define a decision variable s_{ij} as

$$s_{ij} = p_{ij} \cdot \beta + (1 - \eta_{ij})a_{ij} + \eta_{ij}b_{ij} + \frac{1}{2}c_{ij},$$

where β is an arbitrary value selected from the control bound set U_i (6). For instance, it could represent maintaining the current velocity (i.e., zero acceleration input, $\beta = 0_n$). The

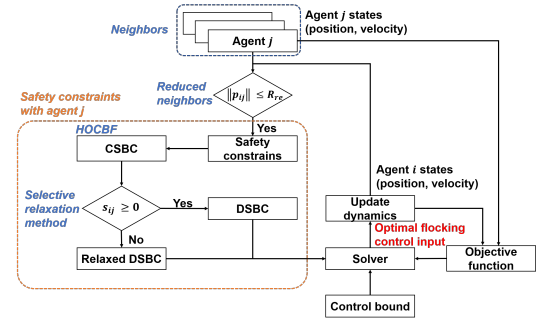


Fig. 5: The overall process of the proposed framework for the selective relaxation method.

decision variable s_{ij} is obtained by rearranging (18) after replacing u_i with β and ω_{ij} with one. Based on the decision variable s_{ij} , the relaxation method is only used when $s_{ij} < 0$. This means that if the original DSBC (i.e., (18) with $\omega_{ij} = 1$) does not have β as a control input, then the relaxation variable is utilized to expand the DSBC. Note that, β is always included in all DSBCs for $s_{ij} \geq 0$. Thus, the intersection of these DSBCs could have some elements because β is always a part of it. This is illustrated in Fig. 4(a). On the other hand, when $s_{ij} < 0$, the relaxation method should be used to expand the DSBC towards β . This implies that the relaxed DSBC can contain β by increasing ω_{ij} , as shown in Figs. 4(b) and (c). Therefore, based on the proposed strategy, all DSBCs can contain states around β , leading to the non-empty admissible control space.

Based on the value of s_{ij} , the relaxation variable is selectively applied to f_{ij} in (12) as

$$f_{ij} = \begin{cases} (1 - \eta_{ij})a_{ij} + \eta_{ij}b_{ij} + \frac{1}{2}c_{ij} & \text{if } s_{ij} \geq 0, \\ (1 - \eta_{ij})\omega_{ij}a_{ij} + \eta_{ij}b_{ij} + \frac{1}{2}c_{ij} & \text{otherwise.} \end{cases} \quad (19)$$

Note that the number of optimization variables is now reduced to $n + |N_i^s|$ where $N_i^s = \{j \in N_i^r | s_{ij} < 0\}$. With the proposed selective relaxation method, the optimal flocking control input for agent i is generated as

$$(u_i^*, \omega_i^*) = \underset{u_i \in \mathbb{R}^n, \omega_i \in \mathbb{R}^{N_i^s}}{\operatorname{argmin}} J_i + K_\omega \|\omega_i - \hat{\omega}_i\|^2, \\ \text{s.t. } -p_{ij} \cdot u_i \leq f_{ij}, \\ \|u_i\| \leq u_{max},$$

where f_{ij} is from (19), K_ω is the weight for the cost of relaxation, $\omega_i = [\omega_{ij_1}, \dots, \omega_{ij_{|N_i^s|}}]^T$, $j_k \in N_i^s$, and $\hat{\omega}_i = 1_{N_i^s} \in \mathbb{R}^{N_i^s}$ with its element being one, representing the nominal case of ω_i , i.e., no relaxation. The selection of β serves as a candidate control input and can be a strategy for determining the action an agent should take in situations of infeasibility. Two possible strategies for selecting β are: choosing a control input that stops the agent or choosing a β that maintains the current state. Setting β as $-u_{max}v_i/\|v_i\|$ to stop the agent in infeasibility situations can help avoid collisions between agents, as shown in [16], where maximum braking is applied in such cases. On the other hand, selecting β to maintain the current velocity, which was determined as

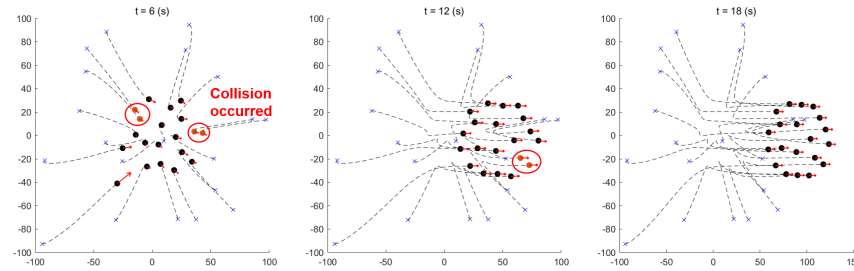


Fig. 6: Trajectory of the ACS flocking algorithm.

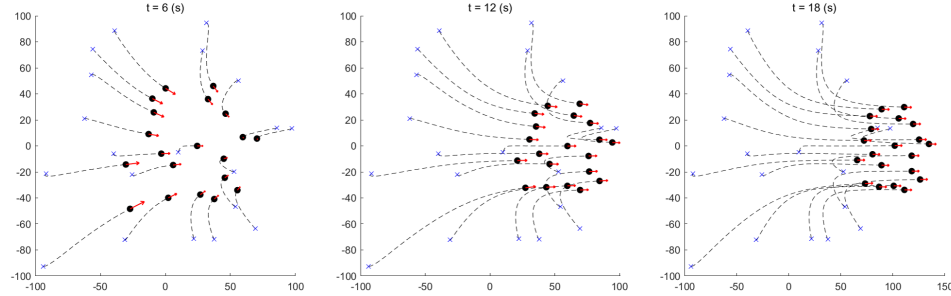


Fig. 7: Trajectory of the safety-critical optimal flocking algorithm.

the optimal solution in the previous step, can also be a good strategy. The overall process of the proposed framework is shown in Fig. 5.

VI. NUMERICAL SIMULATIONS

In this section, we investigate the superiority of the proposed optimal flocking algorithm compared with the existing rule-based flocking algorithm [23], [24] through numerical simulations. In addition, compared with the existing equal distribution method [16]–[18] and relaxation method [16], [19], the validity of the proposed feasibility enhancement methods is demonstrated with improved feasibility and computation efficiency. In the simulation environment, the initial states are randomly selected while satisfying $\mathbf{x}(0) \in \bar{\mathcal{C}}$ and the MAS is assumed to be fully-connected, which means that communication range R_c is infinite. Other parameters are listed in Table I. All computations are performed on 2.5 GHz Intel Core i9 processor using the MATLAB 2021b *fmincon* solver.

TABLE I: Simulation parameters

Parameter	Value	Unit
Dimension, n	2	
Sampling time, Δt	0.1	s
Terminal time, t_f	30	s
Maximum speed, v_{max}	15	m/s
Maximum magnitude of control input, u_{max}	10	m/s ²
Safety distance, D_s	10	m/s
Desired velocity, v_d	(7.5, 0)	m/s
Objective function weights, $(\lambda_1, \lambda_2, \lambda_3)$	(3, 2, 5)	

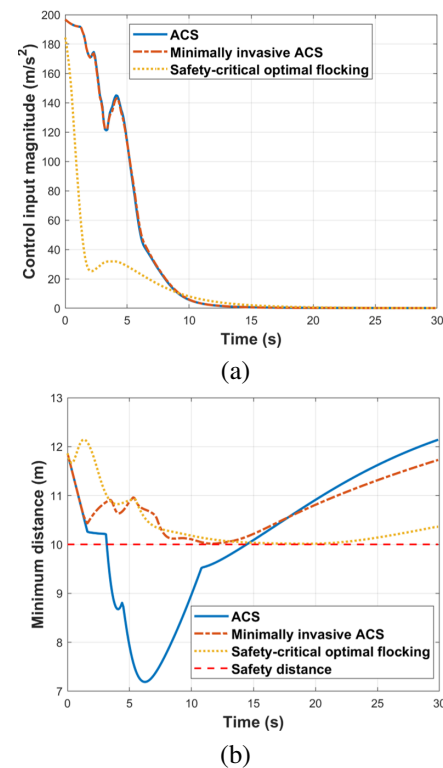


Fig. 8: Comparison of flocking algorithm performances. (a) Control input magnitude. (b) Minimum distance.

A. Comparison of rules-based flocking and safety-critical optimal flocking

First, the performance of the rule-based flocking algorithm and the proposed safety-critical distributed optimal flocking

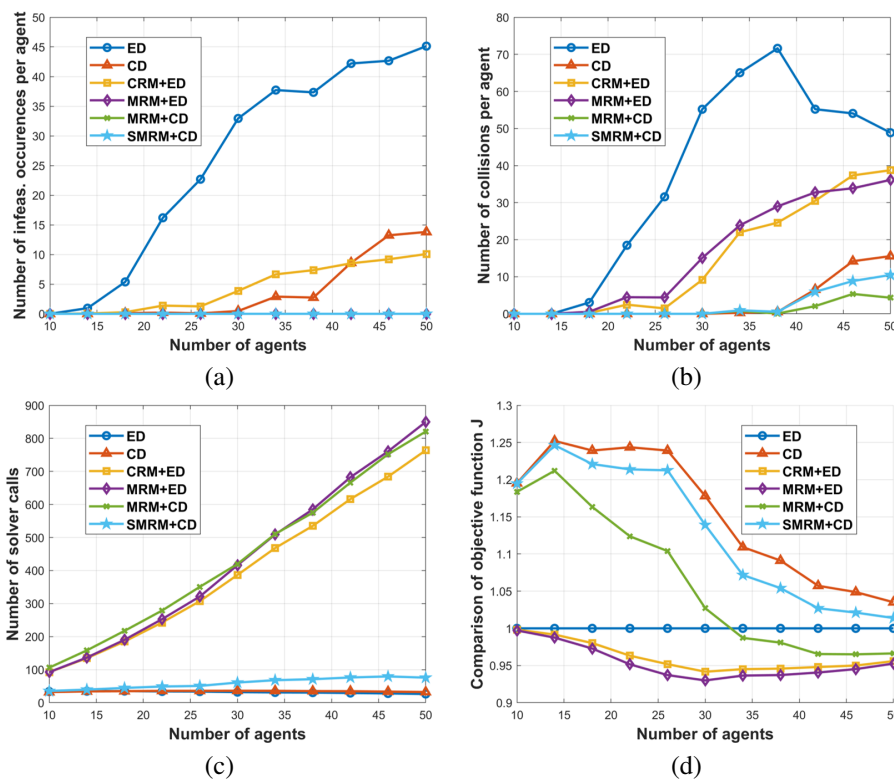


Fig. 9: Comparison of performances among different HOCBF techniques. (a) Infeasibilities. (b) Collisions. (c) Solver calls. (d) Objective function value.

algorithm using the collision risk distribution method with selective relaxation variables in HOCBF are compared. The number of agents N is 20, $\gamma_1 = \gamma_2 = 2$, $\beta = (0, 0)$, relaxation weight for objective function K_ω is 500, the calculated reduced neighbors radius R_{re} is $64m$ using (16), and the desired relative distance R_d is $44m$. The augmented Cucker-Smale (ACS) algorithm [23], [24] is adopted as a rule-based flocking algorithm, which tries to align the velocity of all agents and maintain the desired relative distance. In addition, to check the effect of the HOCBF, the control input of the ACS is made minimally invasive by the HOCBF. The minimally invasive ACS controller refers to the minimal modification of ACS with the construction of an objective function using a quadratic cost minimizing the difference between the nominal $\hat{u}_i$ (i.e., ACS) and optimal control input u_i while ensuring safety with the HOCBF [16] as:

$$\begin{aligned} u_i^* &= \operatorname{argmin}_{u_i \in \mathbb{R}^n} \|u_i - \hat{u}_i\|^2, \\ \text{s.t. } -p_{ij} \cdot u_i &\leq f_{ij}, \\ \|u_i\| &\leq u_{max}, \end{aligned}$$

For an illustrative comparison, the trajectories of the ACS and the proposed safety-critical optimal flocking are shown in Figs. 6 and 7, respectively. The initial safety constraints among agents, $h_{ij}(0)$ and $\psi_{1,ij}(0)$, are set to be greater than 10. In Fig. 6, there are agents that fail to meet the safety distance and collide with other agents (highlighted in the orange color), whereas in Fig. 7, there is no collision throughout the trajectory.

Figure 8 shows that the ACS uses a large amount of control input to quickly gather agents; however, since it does not consider the safety distance, it causes agents to be closer than the safety distance, resulting in collisions. On the other hand, the minimally invasive ACS achieves flocking without collision, but it still uses a large amount of control input. It can be confirmed that the proposed safety-critical optimal flocking algorithm achieves flocking most effectively without collision, which shows the superior flocking performance.

B. Performance analysis of the HOCBF techniques

Next, the performance of different HOCBF techniques is investigated. There are five techniques to compare: equal distribution (ED), collision risk distribution (CD), conventional relaxation method (CRM) with ED, modified relaxation method (MRM) with ED, MRM with CD, and selective modified relaxation method (SMRM) with CD. Here, ED distributes half of the CSBC for the DSBC as used in [16]–[18], CD considers the relative collision risk as in (14), CRM is the relaxation method used in [16], MRM refers to the relaxation method proposed in Section V-A, and SMRM selectively applies the relaxation method according to the decision variable described in Section V-B. This simulation investigates the number of infeasibility occurrences and collisions while changing the number of agents from 10 to 50 through 10 Monte-Carlo simulations. The parameters used in the simulation are as: $\gamma_1 = \gamma_2 = 0.6$, the calculated reduced neighbors radius R_{re} is $294m$, the desired relative distance R_d is $10\sqrt{N}$, β is $(0, 0)$, and relaxation weight for objective function K_ω is 500. The

initial safety constraints among agents, $h_{ij}(0)$ and $\psi_{1,ij}(0)$, are set to be greater than 10. Figure 9(a) shows the average number of infeasibility occurrences for each agent. As shown in the figure, the infeasibility of ED increases as the number of agents increases. On the other hand, the feasibility problem of CD begins to occur after 30 agents, which shows significantly improved feasibility compared with ED. In addition, CRM with ED still has feasibility problems, but the proposed MRM and SMRM with CD have no feasibility problems even when the number of agents increases. Figure 9(b) shows the average number of collisions for each agent. Similar to the infeasibility occurrences, ED has the most collisions whereas CD has much fewer collisions. It is important to note that in relaxation-based methods, collisions can still occur despite achieving feasibility because the safety constraints have been relaxed. Similar to the previous results, the comparison between MRM with ED and MRM/SMRM with CD reveals that the MRM/SMRM with CD techniques exhibit fewer collisions thanks to the CD effect. Figure 9(c) shows the number of solver calls to compute the control input. Except for the SMRM method, the relaxation-based methods increase the number of solver calls as the number of agents increases. On the other hand, SMRM has a similar number of solver calls to those of ED and CD. This result shows that the amount of computation for SMRM does not increase even if the number of agents increases by selectively applying relaxation variables. Figure 9(d) shows the average values of the objective function J_i in (7) (normalized with the value of ED) which indicates how well flocking is achieved. Although CD seems to achieve flocking inefficiently due to its larger objective function value compared with ED, this is because agents behave less effectively for flocking in order to avoid collisions. Besides, the value of the objective function is smaller when relaxation is used. This is because the relaxation variables expand the admissible control space to find control inputs that could make the objective function smaller.

VII. CONCLUSIONS AND FUTURE WORK

In this study, we proposed a safety-critical optimal flocking algorithm by formulating the optimal control problem with safety constraints. An objective function is designed to achieve flocking and the HOCBF is designed as safety constraints. The CSBC is first derived by the HOCBF, and the CSBC is distributed to each agent for the distributed operation as the DSBC. In order to effectively deal with the feasibility problem that can occur in the optimal flocking control problem, the DSBC with the collision risk distribution coefficients and selective relaxation variables are proposed. Through simulation results, it is confirmed that the proposed methods can greatly improve feasibility without significant changes in computation. In future work, we will develop a CBF-based flocking controller for a more realistic dynamic model and demonstrate its performance through real-world experiments.

APPENDIX

The quadratic function represented by (17) has two solutions, denoted by r' and r , where $r' < r$. In order to satisfy

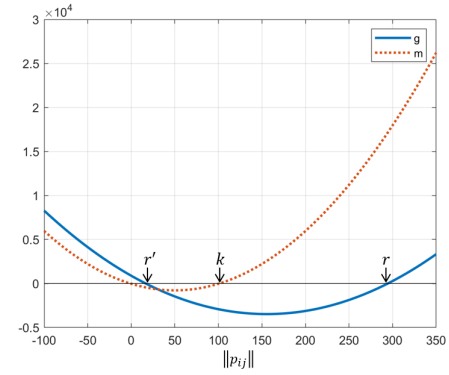


Fig. 10: Relationship between solutions.

the inequality expressed in (17), it is necessary for the norm of p_{ij} to be greater than or equal to r , i.e., $\|p_{ij}\| \geq r$. However, there exists a second solution, $\|p_{ij}\| \leq r'$, which also satisfies the inequality, but this solution does not satisfy the safe sets condition expressed in (3). In order to demonstrate this, let us denote the worst-case scenario for $\psi_{1,ij}$ of C_1 as $m(\|p_{ij}\|)$. We can compute $m(\|p_{ij}\|)$ as

$$m(\|p_{ij}\|) = \frac{\gamma_1}{2} \|p_{ij}\|^2 - 2v_{max} \|p_{ij}\| - \frac{\gamma_1}{2} D_s^2.$$

Considering that the relative distance is greater than zero, the solution for $m(\|p_{ij}\|) \geq 0$ is $\|p_{ij}\| \geq k$ where $k = \frac{2v_{max}}{\gamma_1} + \sqrt{\left(\frac{2v_{max}}{\gamma_1}\right)^2 + D_s^2}$. In other words, to belong to C_1 in the worst-case, $\|p_{ij}\| \geq k$. Furthermore, considering (17), the functions g and m are related by

$$g(\|p_{ij}\|) = l(\|p_{ij}\|) + \gamma_2 m(\|p_{ij}\|),$$

where $l(\|p_{ij}\|) = -2(u_{max} + \gamma_1 v_{max}) \|p_{ij}\| + 4v_{max}^2$. To check the relationship between r' , r and k , substituting k into the above equation gives

$$g(k) = l(k) < 0.$$

From this, we can figure out that $r' < k < r$ as shown in Fig. 10. Therefore, since $\|p_{ij}\| \geq k$, $\|p_{ij}\| \geq r$ is the only range that satisfies safety in the worst-case scenario.

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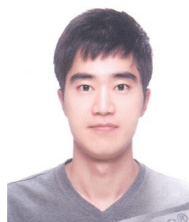
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